DEPARTMENT OF MATHEMATICS

Assignment – Module 1: Group Theory

Course: BITMA401

Introduction to Group Theory

Q.No	Question
1.	Let G be the set of all non-zero real numbers and let $a * b = 7ab$. Show that $(G, *)$ is an abelian group.
2.	Let G be the set of real numbers not equal to -3 and $*$ be defined by $a * b = a + b + ab/3$. Show that $(G, *)$ is an abelian group.
3.	Prove that a group G is abelian if and only if $(ab)^4 = a^4b^4$ for all $a, b \in G$. Also prove that if $x^2 = e$ for every $x \in G$, then G must be abelian.
4.	Let G = { [1, a, b; 0, 1, c; 0, 0, 1] : $a, b, c \in Z$ } under matrix multiplication. Prove that G is a group. Is G abelian? Justify your answer.
5.	Prove that the set of all rational numbers of the form 5^m · 7^n, where m and n are integers, forms an abelian group under multiplication.

Modular Arithmetic, Subgroups & Cyclic Groups

Q.No	Question
1.	Let G be an abelian group and let $S = \{x \in G : x^3 = e\}$. Determine whether S is a subgroup of G. Justify with proof or counterexample.
2.	Let G be a group and let $Z(G) = \{x \in G : xy = yx \text{ for all } y \in G\}$ (the centre of G). Prove that Z(G) is a subgroup of G.
3.	Determine whether the set of all integers modulo 8, i.e., Z_8, forms a group under addition modulo 8. (a) Is it cyclic? (b) Find all generators of Z_8.
4.	Consider the group $(Z_{40}, +)$. Find the order of the elements 5, 8, and 25. Determine the subgroup generated by each of these elements.
5.	Prove that Z_{13}^*, with multiplication modulo 13, is a cyclic group. Compute all its generators.
6.	(i) Let $\lvert a \rvert = 24$ in a group G. Compute the orders of a^3, a^8, and a^{10}. (ii) List the elements of subgroups $\langle 5 \rangle$ and $\langle 10 \rangle$ in Z_{30}. How many subgroups does Z_{30} have? List a generator for each.

Homomorphisms and Isomorphisms

Q.No	Question
1.	Let $G = (\mathbb{R}^*, \cdot)$ and $H = (\mathbb{R}, +)$. Define $\phi : \mathbb{R}^* \to \mathbb{R}$ by $\phi(x) = \ln\|x\|$. Prove that ϕ is a group homomorphism. Find $\ker(\phi)$ and determine whether ϕ is an isomorphism.
2.	Let $f : G \to H$ and $g : H \to K$ be group homomorphisms. Prove that the composition $g \circ f : G \to K$ is also a homomorphism. Further, if $a \in G$ with $\|a\| = n$ and $\|f(a)\| = k$, prove that k divides n.
3.	Let $G = (\mathbb{R}, +)$ and $H = \{z \in \mathbb{C} : \|z\| = 1\}$ under multiplication. Define $\phi : \mathbb{R} \to H$ by $\phi(t) = e^{(2\pi it)}$. Show that ϕ is a surjective homomorphism, find its kernel.
4.	Let $G = (\mathbb{R}^*, \cdot)$ and consider the map $\phi : G \to G$ defined by $\phi(x) = x^2$. Show that ϕ is a homomorphism. Find $\ker(\phi)$, determine the image of ϕ, and decide whether ϕ is an isomorphism.